

1. Project Types and their Specific Features

Students may complete their term projects in two formats: as a research project or as a software project. A research project may encompass theoretical research (formulation and demonstrating proof of new hypotheses and theories), comparative analysis of existing methods for solving a specific problem, and the development or improvement of methods for solving applied problems. Software projects are typically focused on developing or modifying software solutions.

It is important that a project of either type must demonstrate some degree of innovation and/or benefit, for example:

- proposing new methods or models;
- testing the applicability of existing methods to new tasks where such methods have never been used before;
- analysing and comparing methods that have not been the topic of previous research papers;
- developing a system based on already known methods while solving a specific practical problem that has not previously been solved in this way;
- developing a library that differs significantly from existing analogues (at least from analogues available in the public domain).

Thus, the direct reproduction of a published article or implementation of a simplified analogue of an existing open source code library would not be good term projects as they offer no innovation and bring no benefit to the community.

Different aspects should be considered depending on the type and topic of a project.

Main focus areas of a research project (new methods or theoretical conclusions):

1. A detailed description of the problem being solved or the theoretical problem under consideration, and informal and formal formulation of the problem;
2. Review of the existing methods and research outcomes, and analysis of their main components and disadvantages, which you will try to improve in the course of your work;
3. A detailed description of the proposed method and its rationale or formulation and proof of the theoretical results obtained;
4. Theoretical analysis of the proposed solution to the problem;
5. Computational experiments for empirical analysis of the proposed solutions, and comparison with the existing analogues.

Main focus areas of a research project (“analytical” work):

1. Description of the field of study, as well as the products, solutions, and technologies subject to analysis;
2. Identification of tasks for analysis and testing/comparison;
3. Description of the chosen analysis technique and the rationale for choosing it;
4. Description of testing tools and the rationale for choosing them;
5. Detailed, high-quality presentation of analytical results;
6. Discussion of the analytical results, identification of advantages and disadvantages, and provision of recommendations.

Main focus areas of a software project:

1. Providing the rationale for the proposed system or a system prototype;
2. Developing a detailed formal description of the system (general architecture, class structure and their interfaces, volumetric characteristics, etc);
3. Developing a description of the system from the user's point of view;
4. Providing a rationale for the optimal selection of solutions, including on the basis of experimental analysis of the complexity of calculations;
5. Comparing the system with already known analogues in terms of functionality, efficiency, and user-friendly parameters;
6. Ability to demonstrate the system in operation and availability of documentation.

2. Structure and Contents of the Project Report

2.1. Text structure

The following report structure is recommended:

1. Title (front) page (in the same language as the main body of the report)
2. Contents
3. Abstract (up to 2,000 characters): If you submit your paper in English, the abstract must be provided both in Russian and English. In fact, the abstract is a summary of your work, from which anyone familiar with the field of study should be able to understand what you did in general terms. It usually describes the statement of the problem and the main outcomes of the work in fairly informal terms.
4. List of keywords: 5–10 words or phrases that describe the contents (in the same language as the main body of the text).
5. Introduction: In general, this is both an informal introduction and a brief description of the project on 2–3 pages. The introduction usually provides a description of the subject area, informally formulates the statement of the problem, underscores its relevance and significance, and informally describes the main outcomes of the work, including their innovative nature and relevance.
If the project has a non-standard structure, its further structure should be described at the end of the Introduction so that the reader understands what to expect next. For team (group) projects, the distribution of tasks among project participants should be provided at the end of the Introduction for such a team (group) project.
6. Overview of bibliographical sources: Brief description and characteristics of relevant papers. For research projects: positioning your paper among other contemporary papers (e.g., the proposed method is more effective than the one proposed in the paper [1] because this paper examines an additional case that is not studied in [2], etc). For software projects: review of similar software solutions, their comparative analysis and explanations why they cannot be used to solve the problem. An overview of bibliographical sources should not look like a list of relevant papers, but should provide an analysis of the various papers and position your work as it compares to other papers.
7. Chapters (usually a report consists of 2–5 chapters): The structure depends on the topic of the project. For example:
 - a paper proposing a new method for solving a problem may contain the following chapters: formal formulation of the problem and analysis of its features, description of the proposed method, theoretical analysis of the method, experimental research and comparison with analogues;
 - a paper exploring specific aspects of using a certain method for various tasks may contain the following chapters: description of the method, review of the applicability of the method to various tasks with a description of the relevant tasks,

analysis and rationale for the choice of specific tasks for your research, experimental analysis of the applicability of the method to Task 1 in comparison with analogues, and the same for Task 2, etc;

- a paper devoted to the development of a software system for solving a practical problem may contain the following chapters: description and rationale for all selected architectural solutions, algorithms, and technologies, a description of the approaches taken to testing the proposed solution and the rationale for the selected quality metrics, and the results of testing of the developed system and comparison with known analogues.

Each chapter should end with a summary of the main points, where appropriate. This will help highlight the main idea or results presented in the chapter and ensure a smooth transition to the next chapter.

8. Conclusion: Listing and describing the work outcomes (both positive and negative, if any), as well as prospects for further research.
9. Bibliography (make sure to observe the formatting requirements).
10. Annexes (if any): Annexes should be numbered and listed in the Table of Contents. Standard annexes include: a terminological dictionary (glossary) in the subject area; a list of abbreviations; description of the original data for the experiments; protocols of experiments; and additional results of experiments. Source codes of all software programs developed in the course of the work should not be provided in the annexes because this is perceived as an artificial means of expanding the scope of work. You can include key code fragments if they are necessary to demonstrate original solutions or features of the work.

2.2. Recommendations on the text size and style

A scientific style of writing should be used in the project report. The report should describe all components of your work in detail.

The overall volume of your report depends significantly on the topic of the project. For example, a paper on theoretical informatics, which comprises the formulation and proof of new theoretical results, can take up to 10–15 pages. The “standard” volume of an individual project (both research and software) can be considered approximately 20 pages (20,000 characters). The volume of a team (group) project is greater than the volume of an individual project, given the number of participants.

2.3. Recommendations on the description of the distribution of roles in a team project

In a team (group) project report, it is extremely important to clearly describe each team member’s input. This should be done at the end of the introduction, describing the distribution of tasks between team members. This information should also be repeated at the end of each chapter (except for the Introduction, Bibliography, and Conclusion) to indicate who completed their share of work in the given chapter.

2.4. Recommendations on the description of the applicable project terminology, methods, and outcomes

If readers need to know some basic terms, concepts, and methods in your field of study in order to understand your paper, or if you want to introduce some notation, then you can add a chapter (Description of Fundamental Methods, Definitions, etc) into the body of the paper. This will provide the reader with formal definitions and notations without adding these details to the Introduction.

In the paper, most methods are usually described briefly (e.g., in the bibliographical overview), but the main methods you rely on should be described in detail. For example, if you propose making a modification to existing method X, then a chapter should be dedicated in the body of the paper to providing a more detailed description of this method.

2.5. Recommendations on the description of project outcomes

Keep in mind that the project outcomes will be assessed according to the criteria of relevance, innovation, theoretical significance, practical usefulness, reliability and correctness, and completeness. This means that you need to provide reasons for all of your statements, as well as specifically indicate their innovativeness and relevance, and the potential options for their applicability in practice, etc. Make sure to indicate volumetric characteristics (sample sizes; volume of databases; time spent on experiments; number of analysed options; number of lines of code and code volume in kilobytes, etc) in the experimental part of the paper.

2.6. Recommendations on writing and editing the text

Do not try to write a final version of your text from beginning to end right away. Typically, work on drafting a report text is not a linear process.

1. Determine the basic structure of the text, bearing in mind what main chapters will be featured in the text and in what order. At the same time, it makes sense to think through the logical structure of the Introduction, since it typically provides a brief logical outline of the structure of the paper.
2. Write an overview of your bibliographical sources. In general, this is not where you should start, but it is important that you study the literature and have some understanding of how to position your work in the context of existing materials in the process of your work on the project.
3. Write the main chapters of the paper as a draft version. Perhaps in the process you will realise that the structure of the text needs to be changed; this is normal. Modify the structure of the text and Introduction to make it consistent, coherent, and logical.
4. Write the Introduction and Conclusion.
5. Review the entire text and edit if necessary (the same terms should be used consistently, models and results should be provided after their introduction and brief description, etc), and correct all stylistic errors (eliminating colloquial terms, etc).
6. Review the text for correctness of spelling and grammar. Check the accuracy of the formatting (pictures, necessary captions, correct list of references, etc).

3. Requirements for the Project Report Format

It is highly recommended that you format the report in the LaTeX document typesetting software system. The text will look much more professional, and it is also much more convenient to work with neat structuring of the text, design of mathematical sections and the bibliography. We have prepared course project templates for you in the overleaf online editor, and we recommend working directly in it. Look for the English version here: <https://www.overleaf.com/read/zhznztkhqjkw>.

Everything has already been set up in these templates, with brief descriptions of how to format pictures, tables, formulas, and references to the bibliography provided.

If you still plan to draft your report in Word, or you have not found answers to your questions in the LaTeX template, then carefully study the formatting rules in Annex 7. Please also make sure to use title page templates when formatting your report in Word (Annexes 8–11).

4. Plan of the Term Project

In the middle of the year, students must submit a plan for their term projects. The plan is a document comprised of at least four pages (NOT including the title page, table of contents, abstract, bibliography, and illustrations), which represents the first draft version of your project report. It must be drawn up according to the same rules as for the final document and include fully completed points 1–6 in the report structure provided in section 2.1 (title page, table of contents, abstract, keywords, introduction, overview of bibliographical sources), a plan of further work, and a list of references.

Anticipated outcomes can be described in the abstract and introduction, with a remark provided that these outcomes are so far only planned, but not yet obtained. In terms of further work on a research project, the studied models and methods should be described, providing a specific experimental plan (with a description of data, methods, and comparison tools provided). In a plan for further work on a software project, the project should be broken down into specific components, with a plan for their implementation provided.

A plan for a team (group) project will have a larger scope and will consist of at least four pages plus 2–3 pages for each additional participant. The plan for further work should specify both the general plan of the project, with a provisional breakdown by project participants provided.

If your project involves participants from other teams who are not expected to submit a project plan in the middle of the year, then describe the common part of the project in your plan, indicating your own input. As for other participants' input, just note that it exists, but, if it is critical for understanding the general format of the project, describe it briefly in a couple of sentences.